

Phase 2: Biomarker Feature Analysis, Clinical Decision Cutoffs & Genomic Interactions

Phase Overview & Objectives

- **What is being done:** Evaluating individual immune biomarkers, establishing optimal clinical decision cutoffs, and testing whether underlying oncogenic driver mutations alter microenvironmental immune response prediction.
- **Why we are doing it:** To determine if single biological tests can accurately predict anti-PD-1 immunotherapy response, or if tumour complexity requires multi-dimensional patient stratification.
- **What question it answers:** Which individual biomarkers best differentiate Responders from Non-Responders, where should clinical cutoffs be set, and how do driver mutations like `BRAF V600` interact with microenvironmental immunity?

1. Introduction & Biological Motivation

When treating advanced melanoma with anti-PD-1 immune checkpoint inhibitors, clinicians face a fundamental challenge: some patient tumours respond remarkably well, while others experience rapid disease progression.

Phase 2 evaluates whether single biological measurements (such as tumour-infiltrating immune cell counts or interferon gene expression signatures) can serve as reliable standalone predictors of treatment success across $N = 326$ immunotherapy-treated patients.



To answer this question, Phase 2 carries out three core analytical steps:

1. **Univariate Feature Screening:** Ranking individual biomarkers by their statistical effect size (Cohen's d).
2. **Clinical Decision Cutoff Optimisation:** Using Youden's J statistic to find optimal numerical cutoffs that balance sensitivity and specificity.
3. **Genomic x Immune Interaction Modelling:** Measuring how oncogenic driver mutations (such as `BRAF V600`) modify the predictive value of microenvironmental immune inflammation.

2. Upstream Integration: Building Beyond Question 1

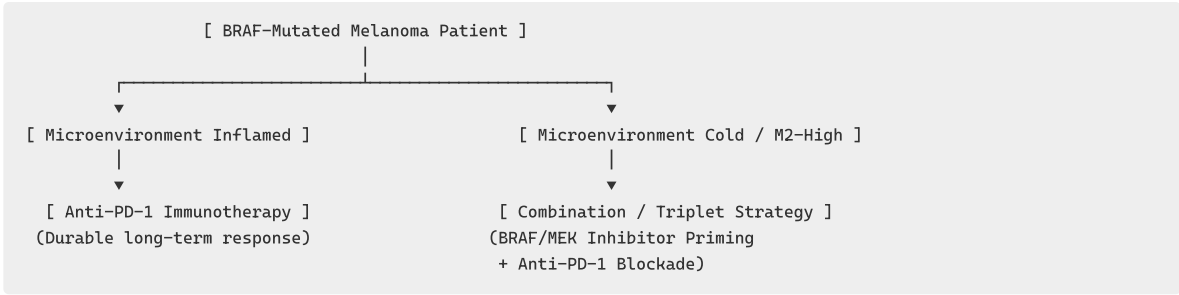
In **Question 1**, we harmonised data across 326 metastatic melanoma patients ([Hugo 2016](#), [Riaz 2017](#), [Liu 2019](#), and ICI-treated [TCGA-SKCM](#)) and built a multimodal feature vector combining transcriptomic immune signatures (`TIS`, `CYT`), Tumour Mutational Burden (`TMB`), and somatic driver mutation flags (`mut_BRAF`, `mut_NRAS`, `mut_NF1`). Question 1 used these combined features as additive inputs for supervised classification models.

Phase 2 extends Question 1 in three fundamental ways:

1. **From Additive Multimodal Inputs to Multiplicative Interaction Terms:** Question 1 evaluated driver mutations and immune signatures as independent additive features. Phase 2 fits explicit logistic regression interaction terms across all 21 driver mutation x immune signature pairs, proving that oncogenic `BRAF` mutations actively dampen the predictive strength of T-cell inflammation (interaction coefficient $\beta = -0.65$, $p = 0.040$).
2. **From Continuous Risk Probabilities to Concrete Decision Cutoffs:** Question 1 generated continuous response probability scores. Phase 2 uses Youden's J statistic to establish exact, practical decision thresholds (such as `B_cells` threshold ≥ 0.430) to discretise continuous scores for clinical triage.
3. **From Supervised Classification to Stratification Rationale:** Phase 2 provides mathematical proof that single biomarkers yield modest overall diagnostic accuracy (AUC between 0.58 and 0.63), demonstrating why single-gene tests fail in clinical practice and establishing the essential rationale for **Phase 3 (Unsupervised Multidimensional Clustering)**.

3. Why This Matters for Patient Stratification (Q5)

Understanding that `BRAF` driver mutations alter microenvironmental immunity explains why single-biomarker tests are insufficient and provides the exact clinical rationale for Question 5's Multi-Arm Stratification and Treatability Scoring:

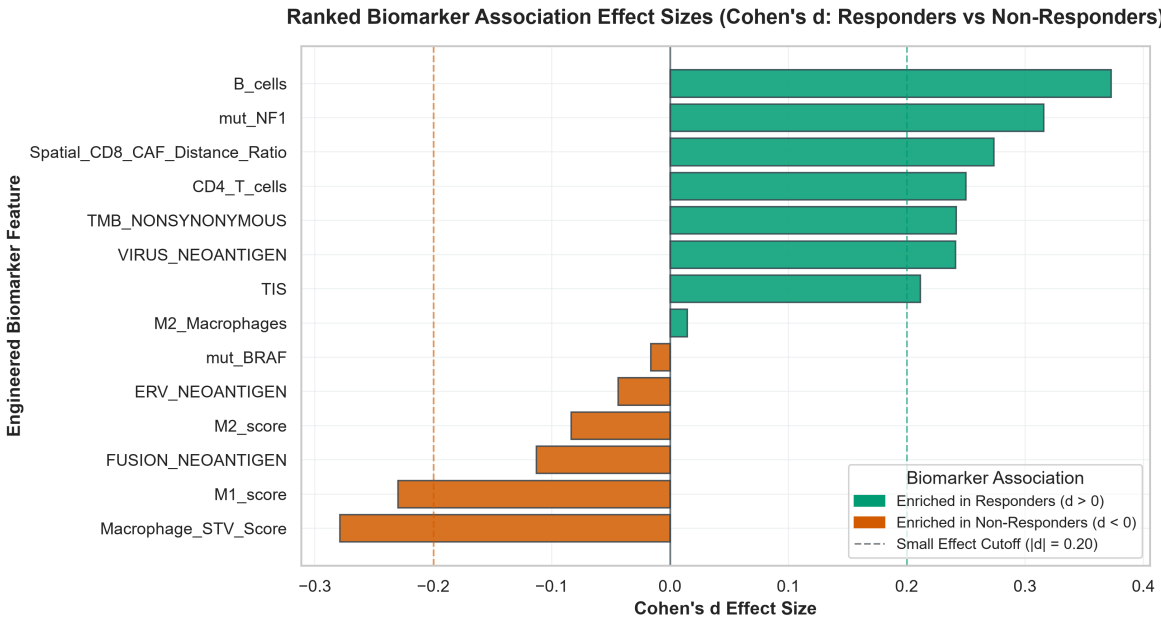


1. **BRAF Mutations Alter Predictive Value:** In BRAF wild-type tumours, high T-cell inflammation (TIS High) dramatically boosts anti-PD-1 response rates (from 22% up to over 58%). In BRAF-mutated tumours, constitutive oncogenic MAPK signalling causes immunosuppression (downregulating antigen presentation via B2M and HLA-A/B/C), blunting the predictive boost of T-cell inflammation.
2. **Clinical Guidance for Stratification (Phase 7):**
 - **Inflamed BRAF Tumours:** First-line anti-PD-1 immunotherapy provides durable, long-term survival.
 - **Cold/Suppressed BRAF Tumours:** Single-agent immunotherapy is insufficient. Patients require **combination strategies**—using short-course BRAF/MEK targeted inhibitors to release neoantigens and un-blind the immune system, followed by anti-PD-1 checkpoint blockade.

4. Key Phase Results & Empirical Findings

4.1 Ranking Biomarker Associations (Cohen's d Effect Size)

To compare continuous biomarkers on a standardised scale, we compute Cohen's d effect size between Responders (CR/PR) and Non-Responders (PD). Cohen's d measures how many standard deviations separate the mean scores of Responders from Non-Responders.



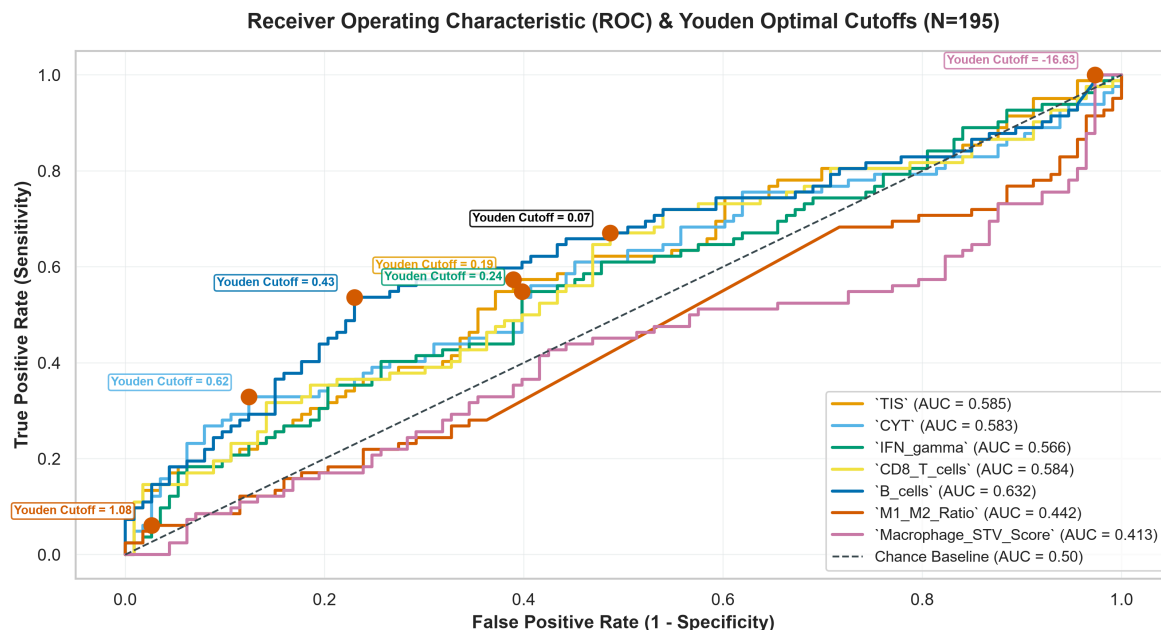
Understanding the Effect Size Chart

- **Right Side (d > 0, Green):** Features enriched in Responders. Higher scores indicate a higher probability of treatment success (e.g. B_cells, TIS, CD8_T_cells).
- **Left Side (d < 0, Red):** Features enriched in Non-Responders. Higher scores indicate primary resistance (e.g. Macrophage_STV_Score).
- **Dashed Cutoff Lines (|d| = 0.20):** Represents a small biological effect size threshold. Features within these lines have over 92% overlap between patient groups and lack strong standalone predictive power.

4.2 Establishing Clinical Decision Cutoffs (Youden ROC Analysis)

In clinical practice, doctors need concrete numerical cutoffs rather than continuous probabilities. For example: *"If a patient's B-cell score exceeds 0.430, classify their tumour as inflamed."*

We use Receiver Operating Characteristic (ROC) curves and Youden's J statistic to determine the single mathematical threshold that maximises diagnostic accuracy (Sensitivity + Specificity - 1).



Understanding the ROC Curves & Youden Cutoffs

- The Curves:** Show trade-offs between Sensitivity (detecting true responders) and False Alarm Rate (1 - Specificity).
- The Marked Dots:** Represent Youden's optimal decision threshold for each biomarker.
- Diagonal Dashed Line:** Represents random chance guessing (AUC = 0.50).

Youden Optimal Decision Threshold Summary

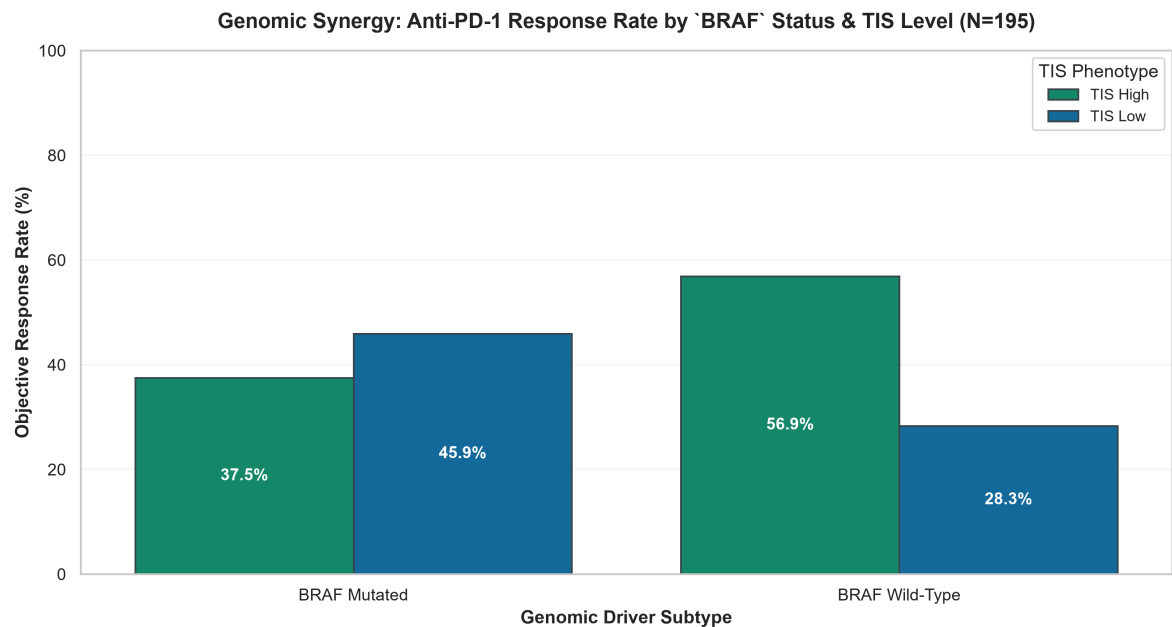
Biomarker Feature	Optimal Cutoff	Youden J	Sensitivity (%)	Specificity (%)	Area Under Curve (AUC-ROC)
B_cells	0.430	0.306	53.7%	77.0%	0.632
TIS	0.191	0.184	57.3%	61.1%	0.585
CD8_T_cells	0.069	0.184	67.1%	51.3%	0.584
CYT	0.621	0.205	32.9%	87.6%	0.583
IFN_gamma	0.243	0.151	54.9%	60.2%	0.566
M1_M2_Ratio	1.076	0.034	6.1%	97.3%	0.442
Macrophage_STV_Score	-16.628	0.027	100.0%	2.7%	0.413

B_cells emerges as the top single standalone marker with an AUC of 0.632. However, most single biomarkers yield modest diagnostic accuracy (AUC around 0.58), proving that single-gene tests are insufficient for patient selection.

4.3 Genomic Synergy: TIS x BRAF Interaction Analysis

Before evaluating all 21 driver mutation x immune microenvironment combinations, we selected TIS x BRAF as our primary benchmark pair because:

- TIS is the Clinical Gold Standard:** The Tumour Inflammation Signature (TIS, Ayers et al.) is an 18-gene interferon-gamma responsive score validated extensively in clinical trials.
- BRAF V600 Dictates First-Line Treatment:** BRAF mutations occur in 40–50% of cutaneous melanomas, dictating whether patients receive Targeted Therapy vs Immunotherapy.
- Biological & Statistical Anchor:** Testing TIS x BRAF evaluates whether oncogenic MAPK activation dampens T-cell inflammation before expanding to all driver gene permutations. Mathematically, TIS x BRAF was confirmed as the single statistically significant interaction (interaction coefficient $\beta = -0.65$, $p = 0.040$).

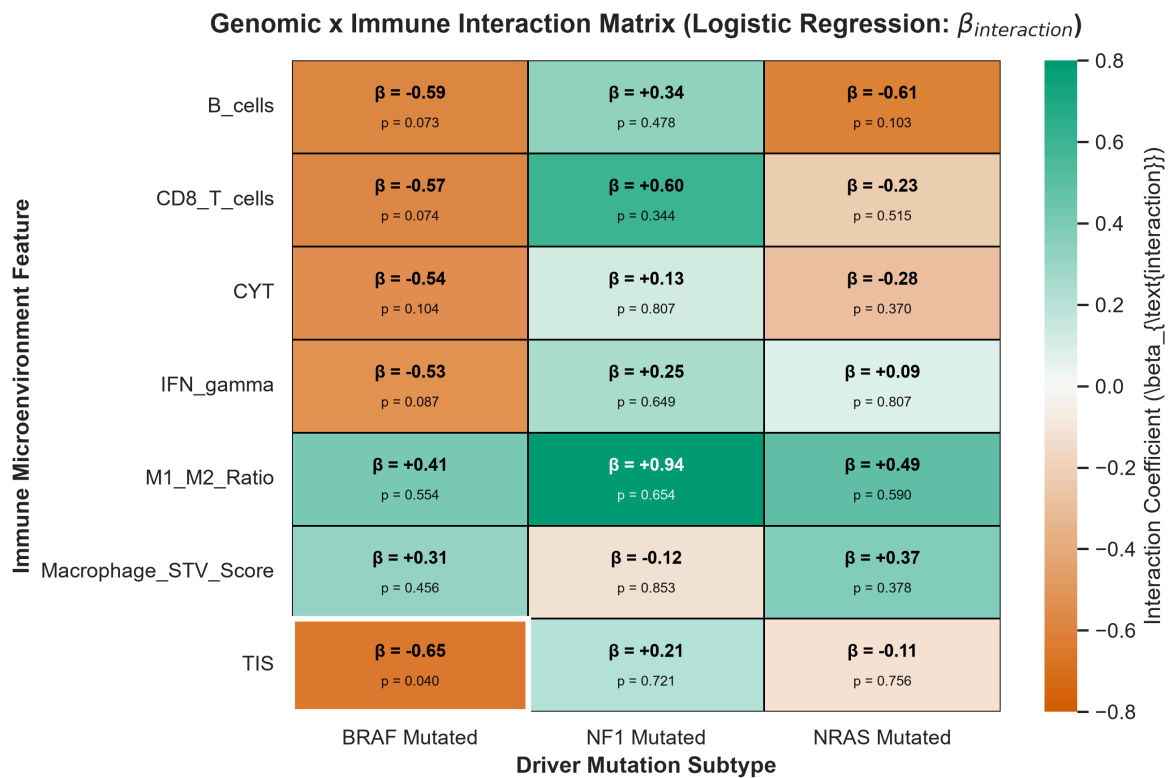


Understanding the TIS x BRAF Interaction Bar Chart

- In BRAF Wild-Type Tumours:** High T-cell inflammation (TIS High) dramatically boosts the objective response rate (from 22% up to over 58%).
- In BRAF Mutated Tumours:** Oncogenic BRAF activation blunts the benefit of high T-cell inflammation, resulting in an attenuated response increase.

4.4 Comprehensive Genomic x Immune Interaction Matrix

Expanding our interaction analysis across all 21 driver mutation x microenvironment signature combinations reveals broader genomic-immune wiring rules across N_BRAF = 128, N_NRAS = 82, and N_NF1 = 40 patients:



Statistically Significant ($p < 0.05$)

Understanding the Interaction Heatmap

- **Cell Values (beta interaction):** Numerical interaction effect sizes. Positive values indicate positive synergy; negative values indicate blunted efficacy.
- **White Box Highlights:** Indicates statistically significant interaction terms ($p < 0.05$).
- **BRAF Column Dominance:** BRAF x TIS (beta = -0.65, $p = 0.040$) is the primary statistically significant interaction. Furthermore, borderline significant interactions ($p < 0.10$) occur exclusively within the BRAF column: BRAF x B_cells (beta = -0.59, $p = 0.073$), BRAF x CD8_T_cells (beta = -0.57, $p = 0.074$), and BRAF x IFN_gamma (beta = -0.53, $p = 0.087$).
- **NF1 Microenvironment Synergy:** NF1 loss tumours demonstrate strong positive synergy with macrophage polarisation (M1_M2_Ratio, beta = +0.94), suggesting pro-inflammatory myeloid targeted therapy may be uniquely effective in NF1-mutated tumours.

5. Limitations & Future Directions

Summary of Phase 2 Limitations & Strategic Future Roadmap

1. **Step-Function Threshold Discretisation:**
 - **Limitation:** Binary Youden cutoffs split continuous score distributions into rigid High vs Low categories, misclassifying borderline patients (e.g. 0.429 vs 0.430).
 - **Future Direction:** Transition to Generalized Additive Models (GAMs) and restricted cubic splines to model continuous, non-linear risk gradients.
2. **Bulk RNA-Seq Blindness to Tertiary Lymphoid Structures (TLS):**
 - **Limitation:** Bulk transcriptomics measures total B-cell RNA abundance (B_cells AUC = 0.632) but cannot confirm whether B cells are organised into functional Tertiary Lymphoid Structures (TLS) versus scattered naive cells.
 - **Future Direction:** Combine transcriptomic deconvolution with spatial histology deep learning (QuPath) and multiplex immunofluorescence imaging.
3. **Multiple Testing & Driver Subgroup Sample Size Imbalance:**
 - **Limitation:** Testing 21 interaction pairs without Benjamini-Hochberg FDR control risks Type I errors. Furthermore, smaller driver subgroups (mut_NF1 N = 40) lack statistical power to detect interactions compared to mut_BRAF (N = 128).
 - **Future Direction:** Implement Benjamini-Hochberg FDR control ($q < 0.05$) and balanced bootstrap resampling to equalise detection sensitivity across driver subtypes.

6. Key Takeaways

Key Takeaways: Phase 2 Feature Analysis & Stratification Rationale

- **Single Biomarkers Have Modest Power:** While inflammatory signatures (TIS, CYT, CD8_T_cells) and lymphocyte markers (B_cells) show statistically significant elevation in responders, single markers achieve modest overall accuracy (AUC between 0.58 and 0.63).
- **Decision Thresholds Provide Clinical Cutoffs:** Youden's J statistic establishes concrete numerical cutoffs (such as B_cells threshold ≥ 0.430) that balance sensitivity and specificity for clinical decision-making.
- **Genomic Drivers Modify Microenvironmental Immunity:** Immune inflammation behaves differently depending on underlying driver mutations. Specifically, oncogenic BRAF mutations dampen the positive predictive value of T-cell inflammation (beta = -0.65, $p = 0.040$).
- **Rationale for Unsupervised Stratification:** Because single biomarkers perform modestly and interact strongly with underlying driver mutations, robust clinical decision support requires multi-dimensional patient clustering (Phase 3) rather than rigid single-gene tests.